# Bring My Cup!

## Personalizing Vision-Language-Action Models with Visual Attentive Prompting

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POSTECH

## 'Cup' is easy. 'My Cup" is hard.



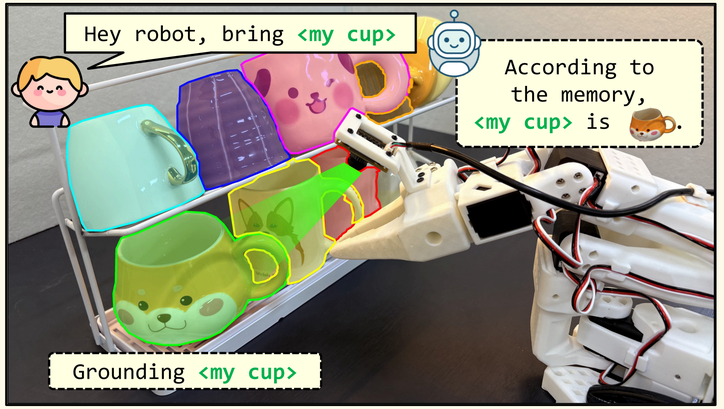

VAP makes 'My' visible for frozen VLAs
Show it → Find it → Mark it → Move it

### 1. Show user's cup

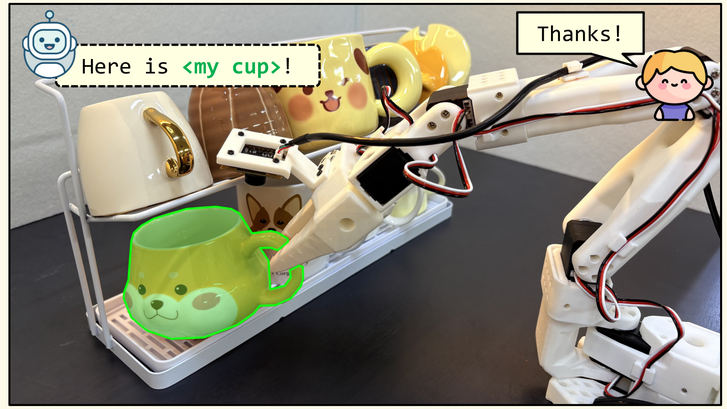


5 images of <my cup>

### 2. Find and mark user's cup using memory

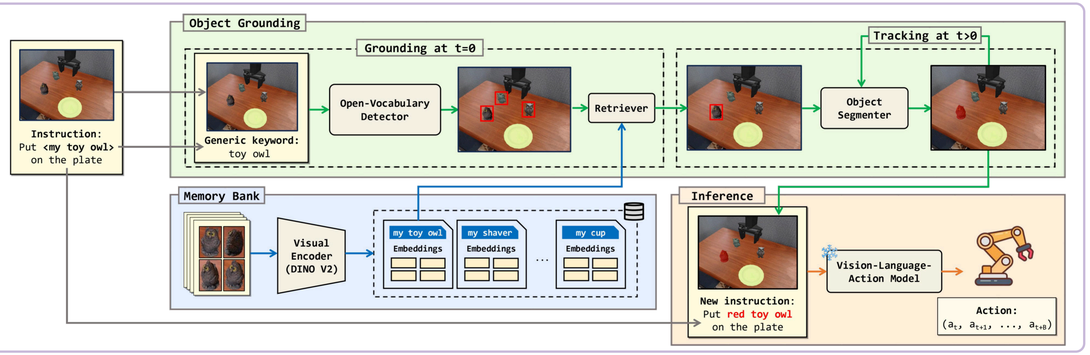



### 3. Bring the user's cup!

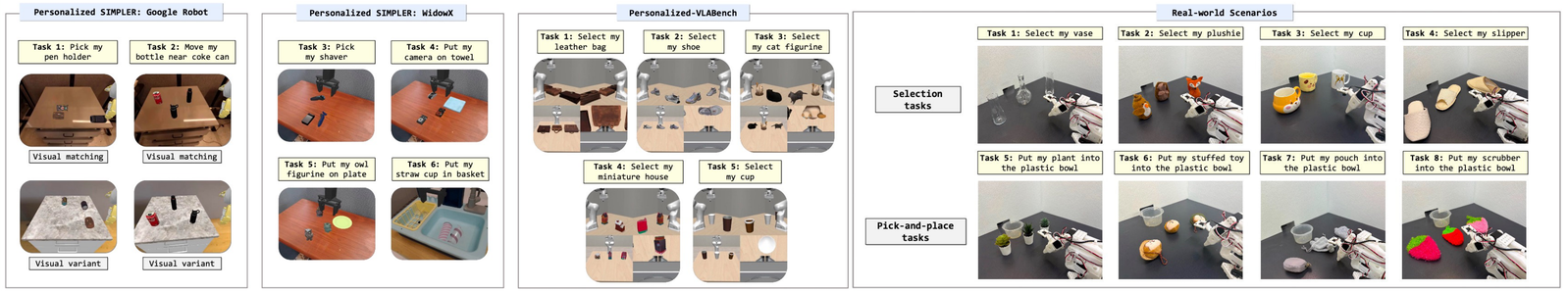



## Overview of VAP

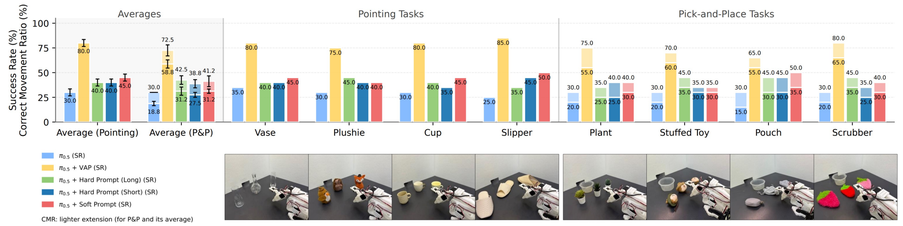
- **VLAs fail at personalized commands** like "my cup."
- Personalization requires **distinguishing a specific object** from look-alike distractors.
- We propose **VAP**, a training-free visual grounding and prompting framework to solve this.




## Benchmarks




## Quantitative Results




**30.0 → 80.0%**
Success rate, on selection

**18.8 → 58.8%**
Success rate, on PnP task

**+ 0.02 s / step**
Little tracking overhead

## Detailed Results

| Method | Task 1: Pick my pen holder | | Task 2: Move my bottle near coke can | |
|---|---|---|---|---|
| | CMR (%) | SR (%) | CMR (%) | SR (%) |
| *Track 1: Visual Matching with Unseen Personalization Objects* | | | | |
| $\pi_0$ | 10.5 | 8.5 | 50.4 | 48.6 |
| $\pi_0$ + Soft Prompt | 33.1 | 24.0 | 60.0 | 51.7 |
| $\pi_0$ + Hard Prompt (Short) | 11.0 | 8.9 | 66.7 | 57.5 |
| $\pi_0$ + Hard Prompt (Long) | 11.2 | 9.1 | 64.5 | 58.8 |
| $\pi_0$ + VAP | **89.2** | **60.3** | **83.2** | **75.0** |
| *Track 2: Variant Aggregation with Unseen Personalization Objects* | | | | |
| $\pi_0$ | 17.4 | 10.2 | 49.2 | 41.3 |
| $\pi_0$ + Soft Prompt | 31.8 | 26.4 | 57.7 | 44.7 |
| $\pi_0$ + Hard Prompt (Short) | 21.6 | 14.4 | 71.5 | 54.2 |
| $\pi_0$ + Hard Prompt (Long) | 17.3 | 12.6 | 67.7 | 51.3 |
| $\pi_0$ + VAP | **87.3** | **58.2** | **72.4** | **62.5** |

| Method | Task 3: Pick shaver | | Task 4: Put camera | | Task 5: Put owl | | Task 6: Put straw cup | |
|---|---|---|---|---|---|---|---|---|
| | CMR (%) | SR (%) | CMR (%) | SR (%) | CMR (%) | SR (%) | CMR (%) | SR (%) |
| $\pi_0$ | 0.0 | 0.0 | 34.2 | 31.7 | 35.2 | 30.3 | 80.6 | 27.8 |
| $\pi_0$ + Soft Prompt | 25.0 | 8.3 | 54.2 | 41.7 | 79.2 | 62.5 | 87.5 | 29.2 |
| $\pi_0$ + Hard Prompt (Short) | 0.0 | 0.0 | 50.0 | 25.0 | 37.5 | 33.3 | 83.3 | 29.2 |
| $\pi_0$ + Hard Prompt (Long) | 0.0 | 0.0 | 53.2 | 27.9 | 36.0 | 31.6 | 83.4 | 30.1 |
| $\pi_0$ + VAP | **82.9** | **71.3** | **92.1** | **92.1** | **100.0** | **95.0** | **100.0** | **75.6** |

| Method | Task 1: Leather bag | Task 2: Shoe | Task 3: Cat figurine | Task 4: Miniature house | Task 5: Cup |
|---|---|---|---|---|---|
| $\pi_{0.5}$ | 33.6 | 30.4 | 35.2 | 29.6 | 40.4 |
| $\pi_{0.5}$ + Soft Prompt | 51.2 | 40.8 | 44.4 | 39.6 | 54.8 |
| $\pi_{0.5}$ + Hard Prompt (Short) | 52.4 | 37.6 | 44.0 | 38.8 | 51.6 |
| $\pi_{0.5}$ + Hard Prompt (Long) | 52.0 | 39.6 | 42.8 | 39.6 | 50.0 |
| $\pi_{0.5}$ + VAP | **89.2** | **54.0** | **51.6** | **52.4** | **60.8** |